



Object Detection

27 October
2025

v. 2.1



Seminar mixed with theory & practical sessions



Theory Journey – How the course builds conceptually

From Vision to Intelligence:

How machines learn to see

Recognition & Tracking:

Teaching machines to follow what they see

Deep Learning for Vision:

Neural networks that understand images

Generative & Responsible Vision:

When machines start to create — and deceive

Applied & Creative Vision: **Designing with AI & looking ahead**

The Practical Journey – Creating the Pixel Machine Lab

Image Manipulation with OpenCV · Mini Image Gallery

Face & Object Tracking · Dataset Annotation

Training Custom Models · Gesture Recognition · YOLOv9



Micro Deep Dives: Generative Vision · Diffusion · Ethics · Bias · AI Tools

Spatial AI and 3D Point Cloud Exploration with Open3D

27 October
2025





Course Goals

Object Detection

YOLO

Fool the AI Gallery

Seeing an object is not enough, we need to know *where* it is.

Real-time Object Detection, and how a surveillance machine has become a creative tool.

Understand *machine perception* as both powerful and limited.

27 October
2025





Object Detection

27 October
2025





Computer Vision Tasks

Classification



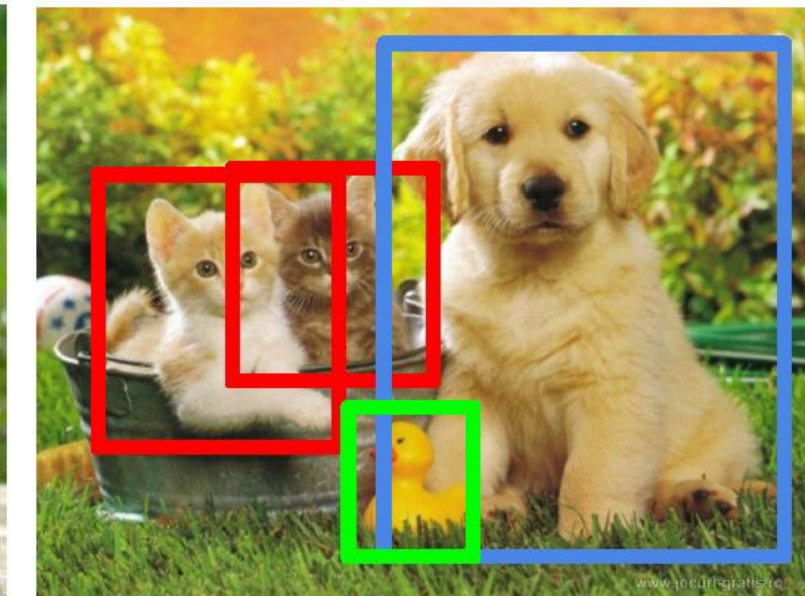
CAT

Classification
+ Localization



CAT

Object Detection



CAT, DOG, DUCK

Instance Segmentation



CAT, DOG, DUCK

Single object

Multiple objects





Computer Vision Tasks

Classification



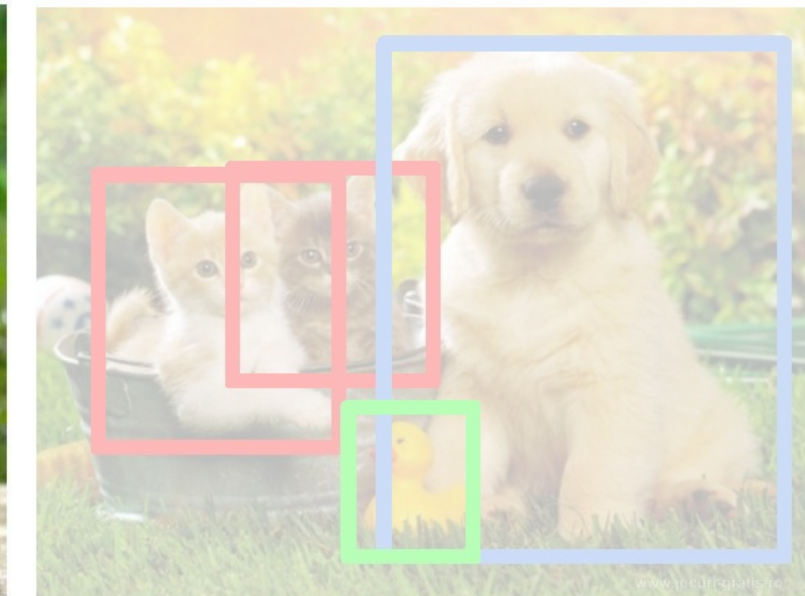
CAT

Classification
+ Localization



CAT

Object Detection



CAT, DOG, DUCK

Instance Segmentation



CAT, DOG, DUCK

Single object

Multiple objects





Classification & Localisation



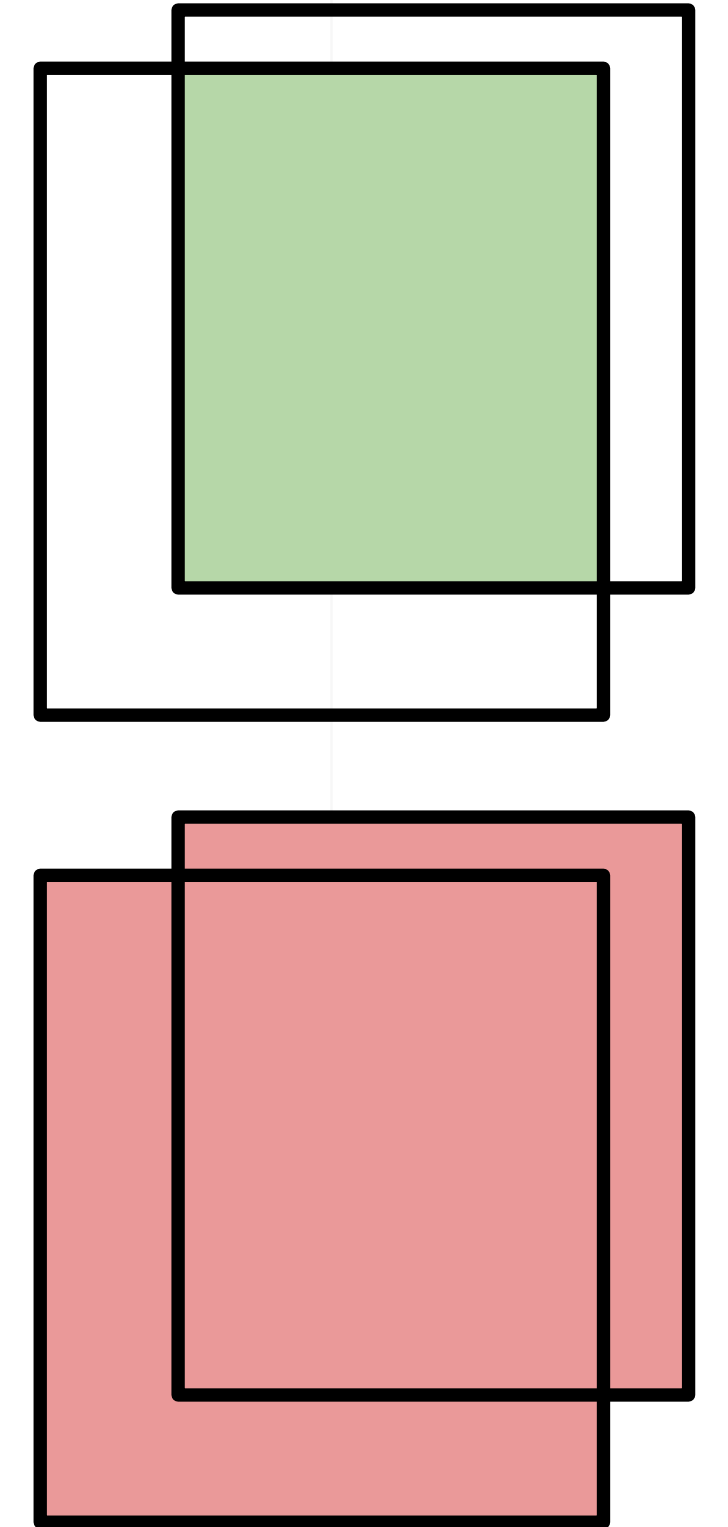
- Classification:
 - Input: Image
 - Output: Class label
 - Loss: Cross entropy (Softmaxlog)
 - Evaluation metric: Accuracy
- Localization:
 - Input: Image
 - Output: Box in the image (x, y, w, h)
 - Loss: L2 Loss (Euclidean distance)
 - Evaluation metric: Intersection over Union
- Classification + Localization:
 - Input: Image
 - Output: Class label + box in the image
 - Loss: Sum of both losses





Intersection Over Union (IoU)

$$\text{IoU}(A,B) = \frac{\text{Intersection}(A,B)}{\text{Union}(A,B)}$$



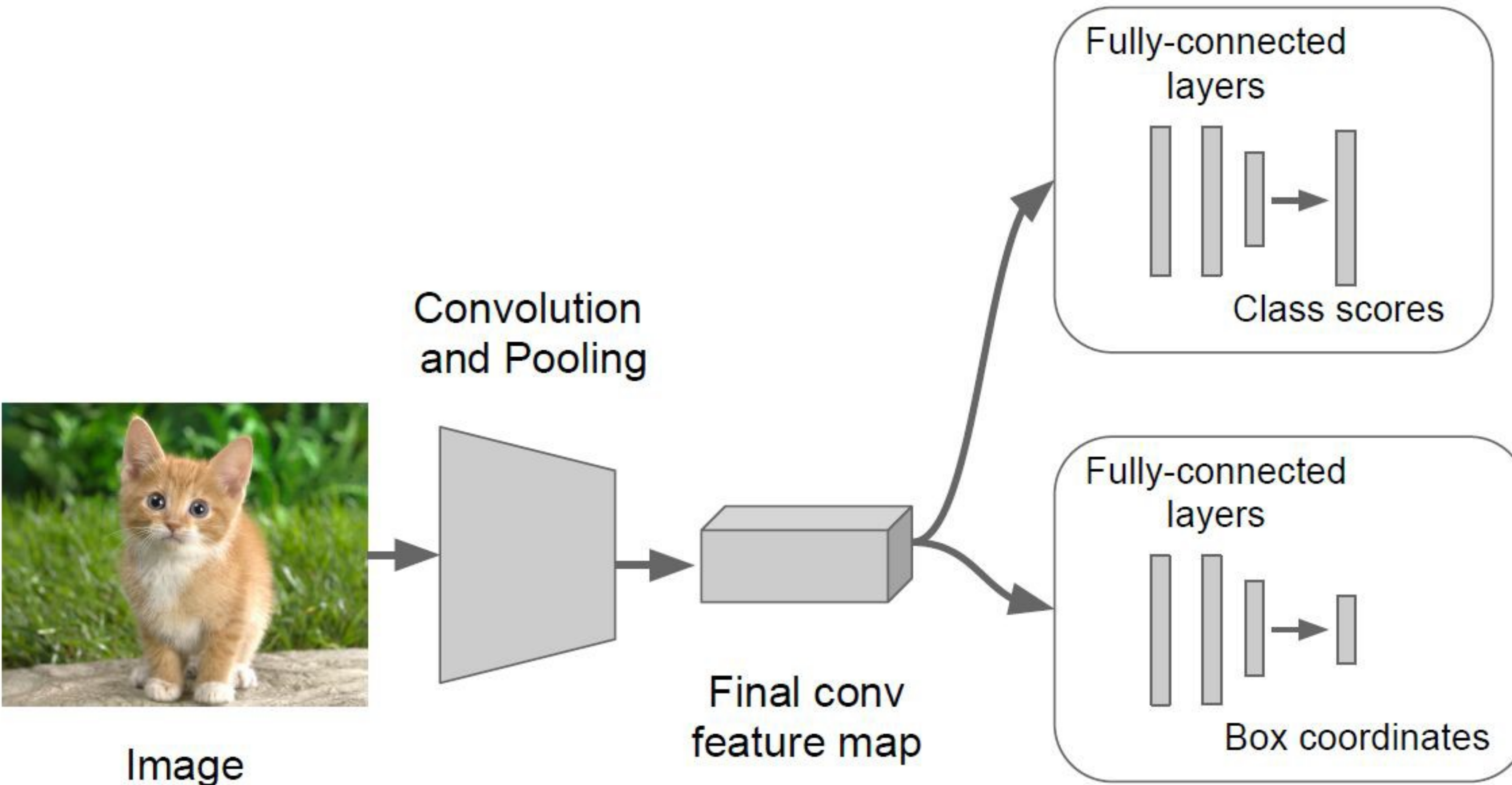
27 October
2025

- Important ' measurement for object localization.
- Used in both training and evaluation.





Classification + Localization: Model



Classification Head:

- C Scores for C classes

Localization Head:

- Class agnostic: (x, y, w, h)
- Class specific: $(x, y, w, h) \times C$



Computer Vision Tasks

Classification



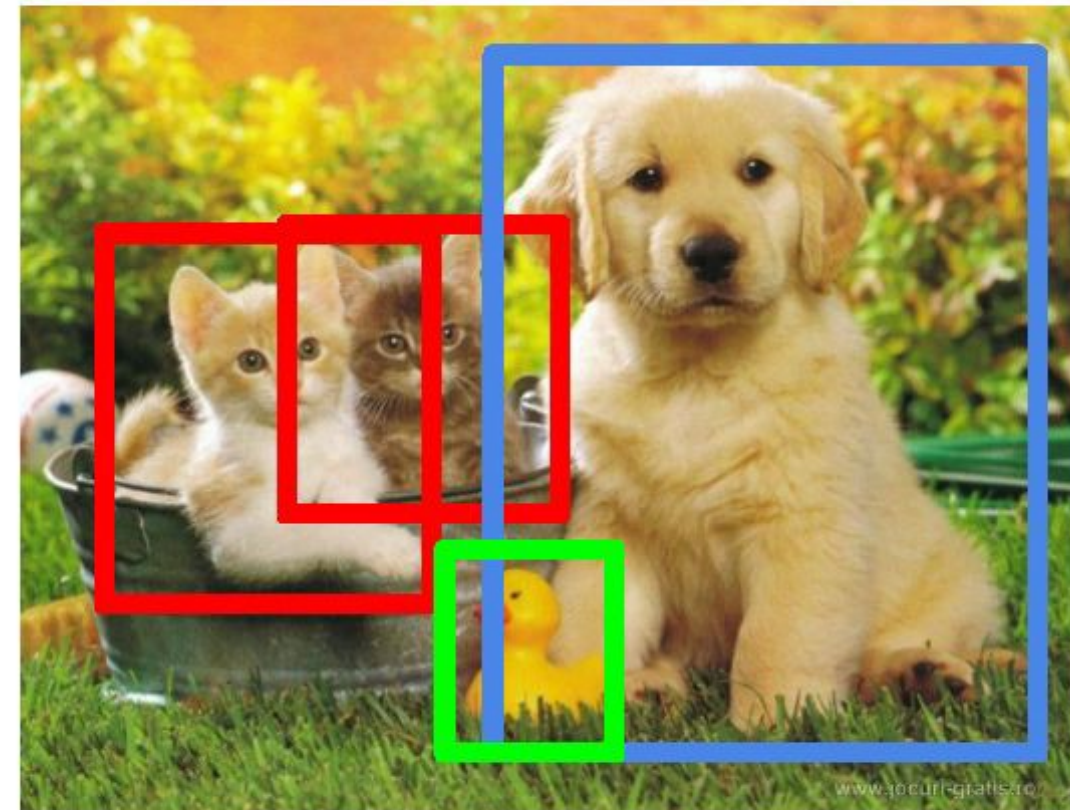
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

**Instance
Segmentation**



CAT, DOG, DUCK

Single object

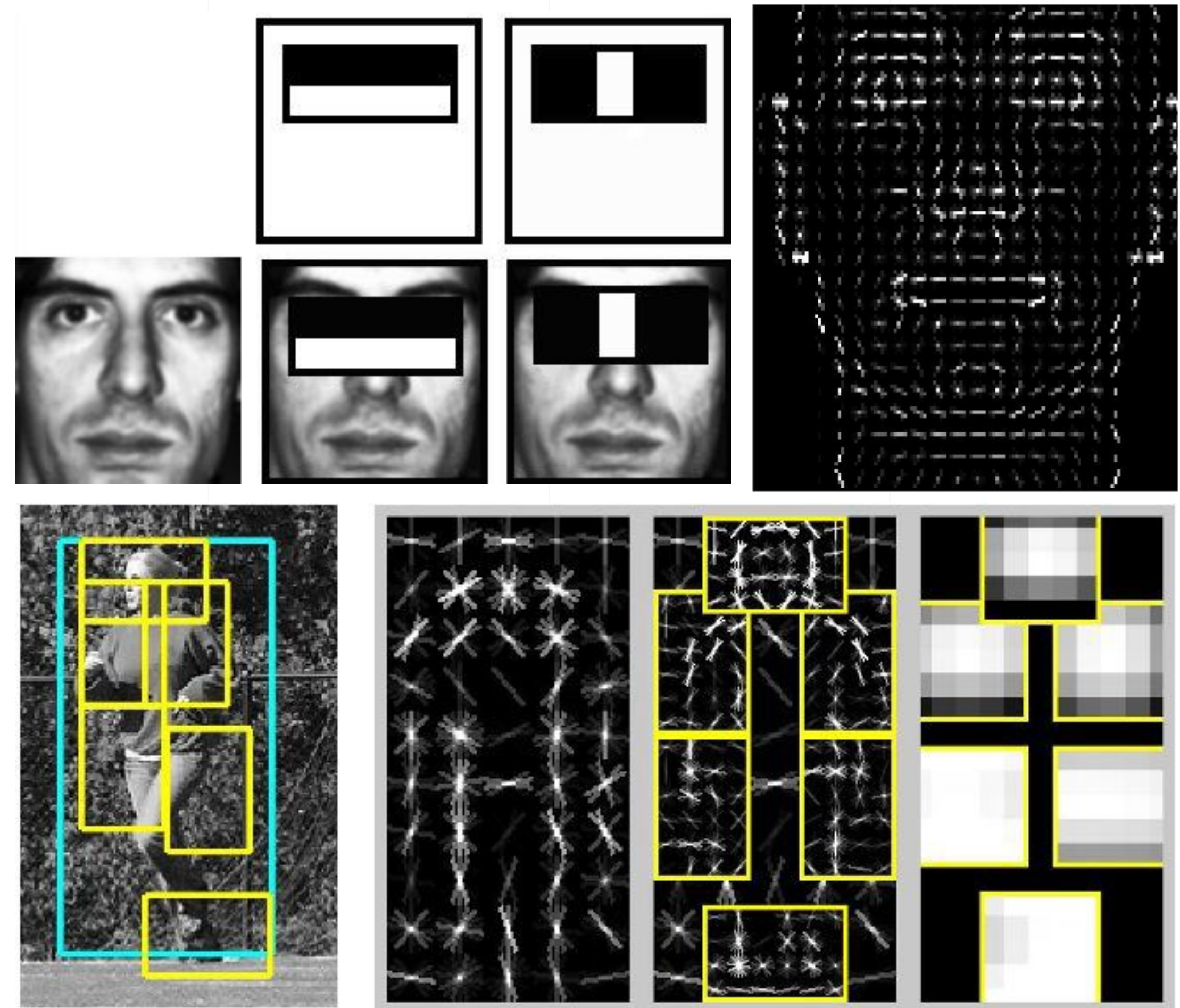
Multiple objects

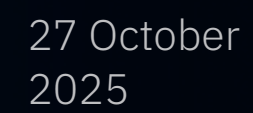


Object Detection 2001-2007

- Rapid Object Detection using a Boosted Cascade of Simple Features (2001)
 - Viola & Jones
- Histograms of Oriented Gradients for Human Detection (2005)
 - Dalal & Triggs
- Object Detection with Discriminatively Trained Part Based Models (2010)
 - Felzenszwalb, Girshick, Ramanan
- Fast Feature Pyramids for Object Detection (2014)
 - Dollar

27 October
2025







Datasets are basically the *teacher* for all vision systems.

Every computer-vision model needs data to learn from.

Humans labeled thousands of images for datasets like COCO — drawing boxes and naming objects.

class.txt -> e.g. Cat, Dog, Car, Bus, Truck, Bird

folder with class images

folder with image labels are bounding Box with object Height, Width, Center (x,y)

27 October
2025

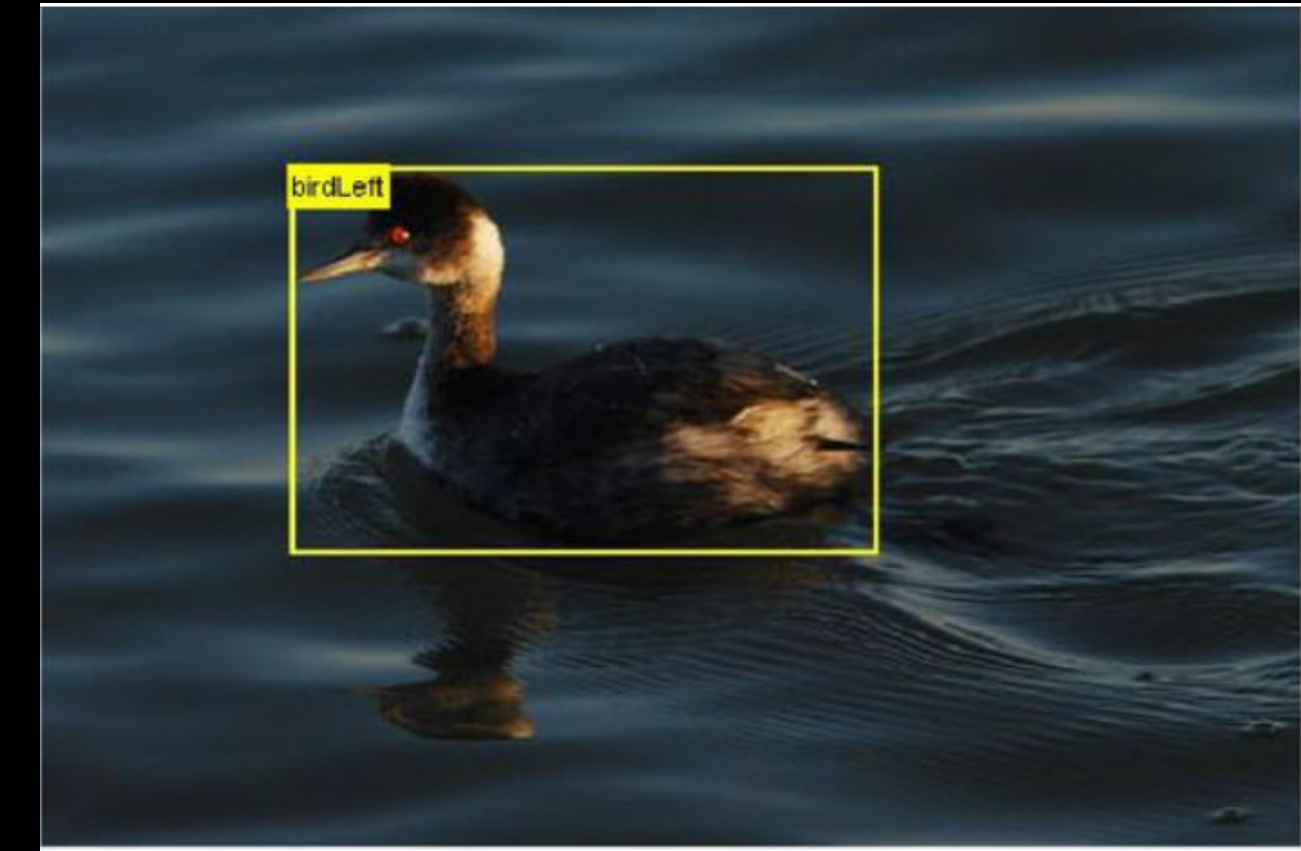
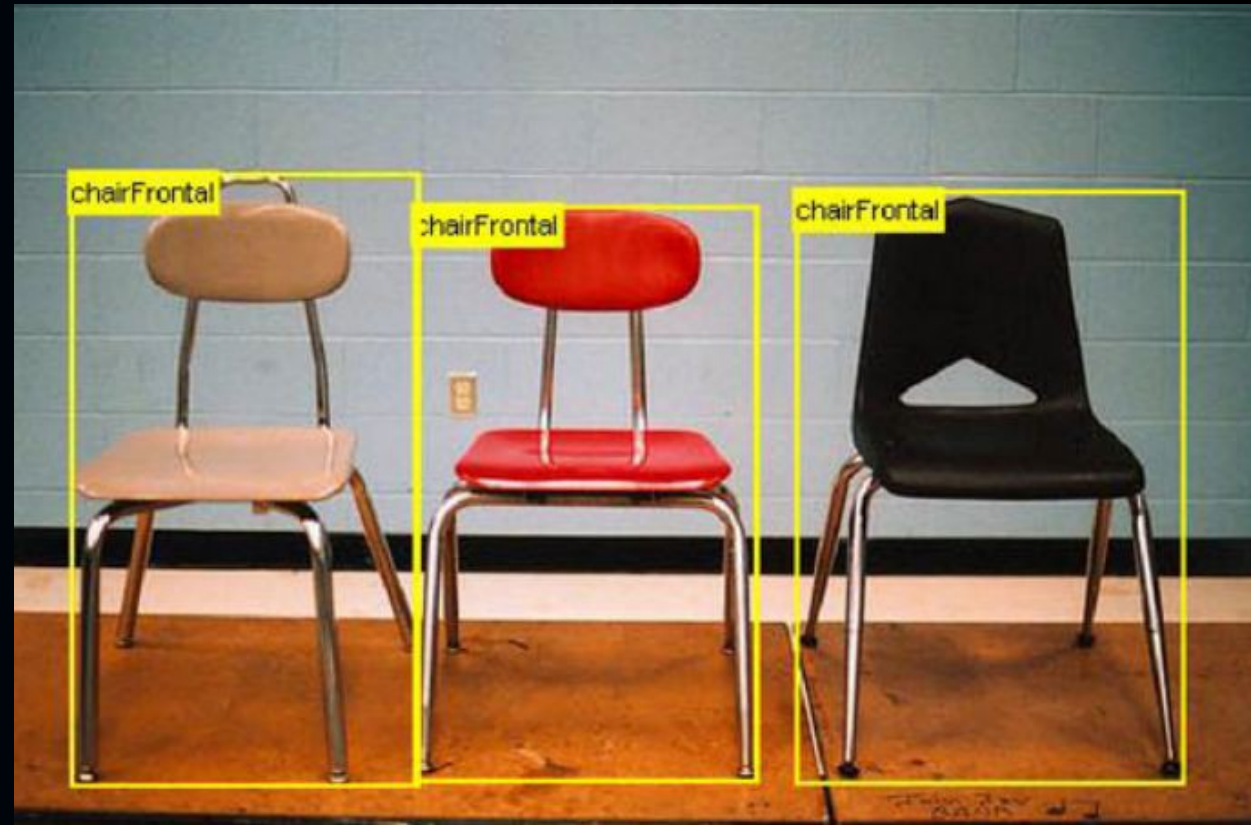


```
└─ yolov8
  ## └─ train
  ##### └─ images (folder including all training images)
  ##### └─ labels (folder including all training labels)
  ## └─ test
  ##### └─ images (folder including all testing images)
  ##### └─ labels (folder including all testing labels)
  ## └─ valid
  ##### └─ images (folder including all testing images)
  ##### └─ labels (folder including all testing labels)
```





Pascal Examples



27 October
2025





COCO Examples



27 October
2025





Every computer-vision model needs data to learn from.
Humans labeled thousands of images for datasets like COCO — drawing boxes and naming objects.
YOLO sees these examples again and again until it learns to predict the boxes itself.
So datasets are basically the *teacher* for all vision systems.

2007

Pascal VOC

- 20 Classes
- 11K Training images
- 27K Training objects

2013

ImageNet ILSVRC

- 200 Classes
- 476K Training images
- 534K Training objects

2015

MS COCO

- 80 Classes
- 328k Training images
- 1.5M Training objects

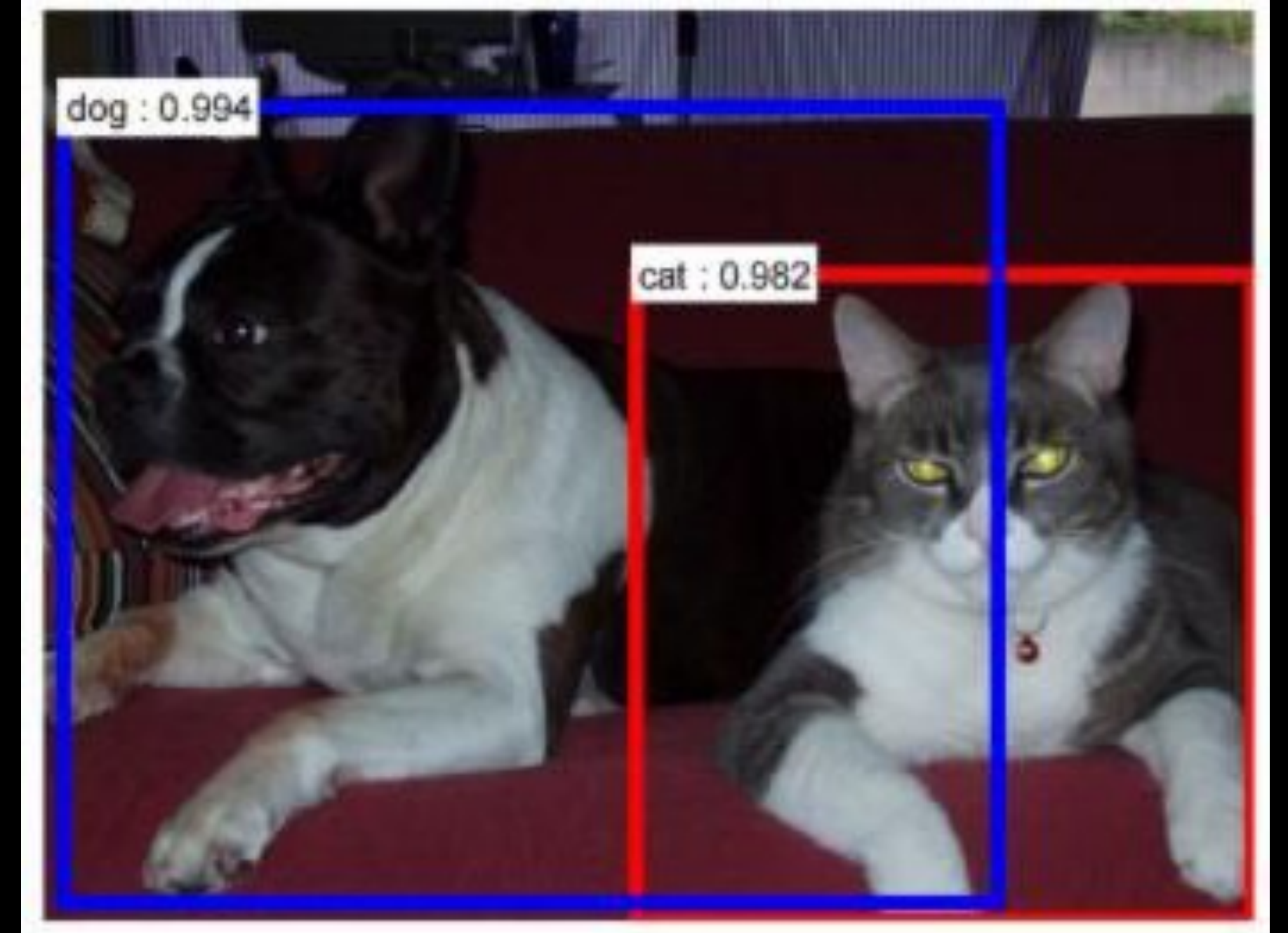
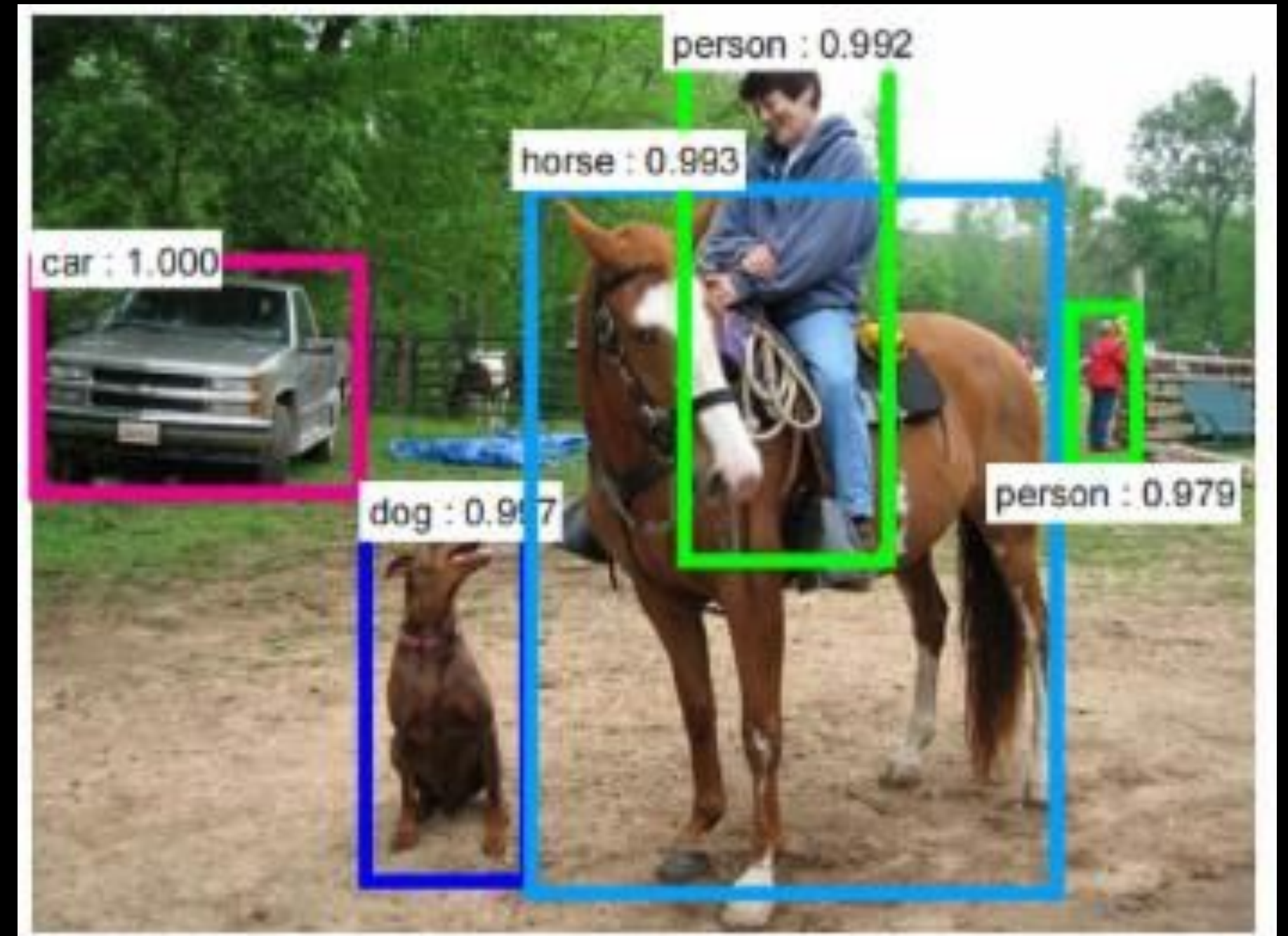




Object Detection

- Input: Image
- Output: For each object class c and each image i , an algorithm returns predicted detections: locations (x,y,h,w) with confidence scores

27 October
2025

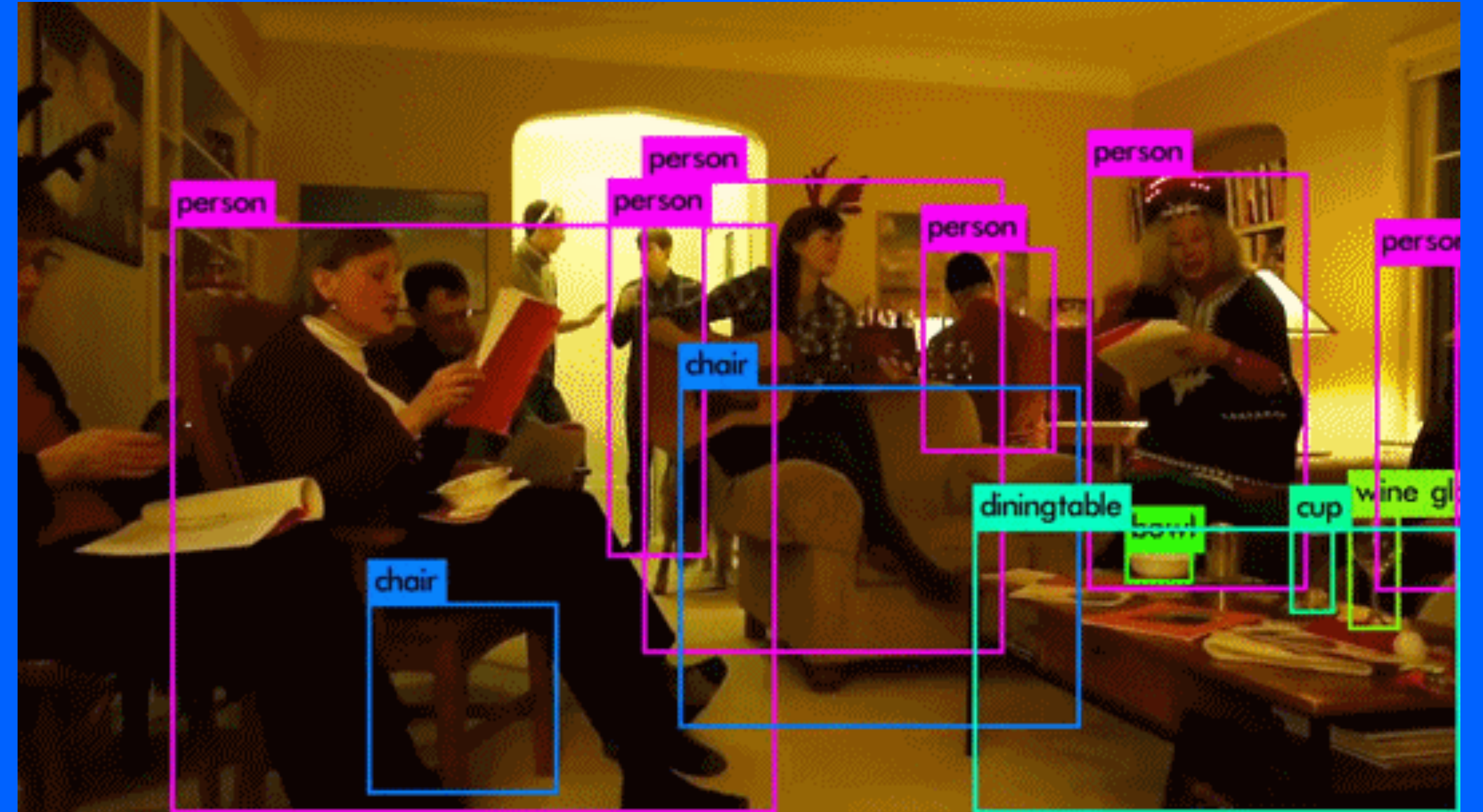




YOLO (You Only Look Once)

Unified Real-Time Object Detection

27 October
2025





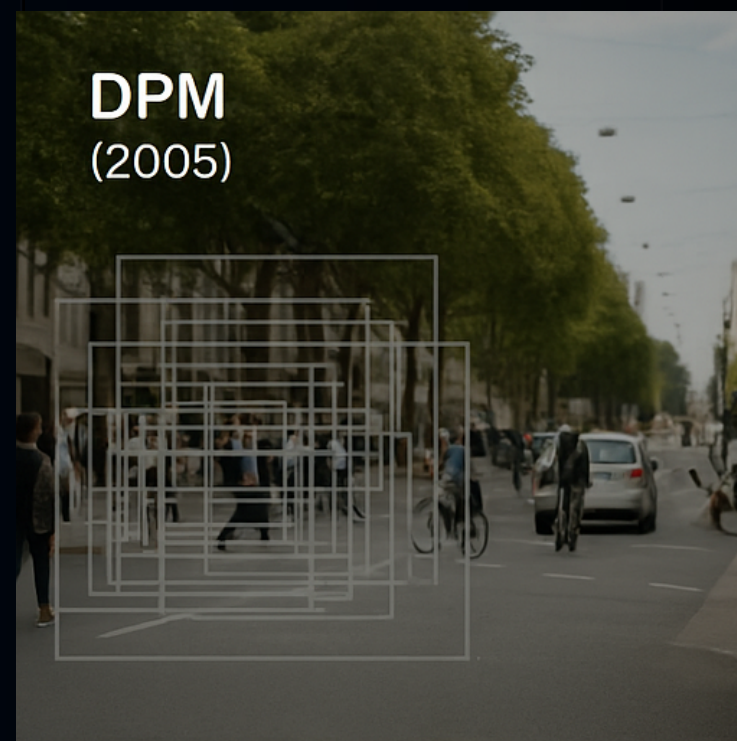
Object Detection evolution over time

DPM

Deformable Parts Model (2005–2010)

🧱 Hand-crafted features (HOG) + sliding windows

🐢 Slow, step-by-step



R-CNN

Region-based CNN (2014)

🧩 Deep features (CNN) + region proposals

⚙️ Faster (with RPN)

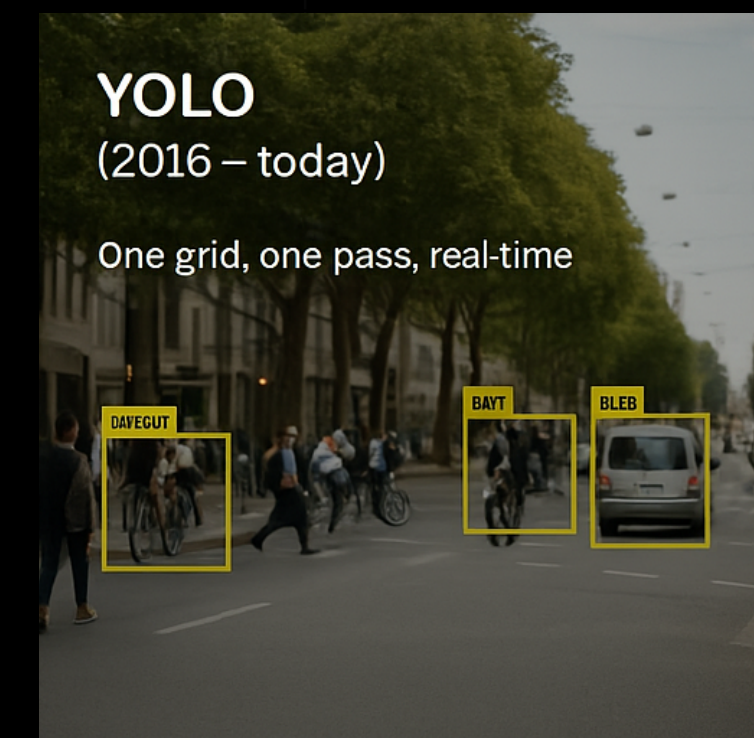


YOLO

You Only Look Once (2016 → today)

⚡ Single CNN, predicts all boxes at once

🚀 Real-time, end-to-end



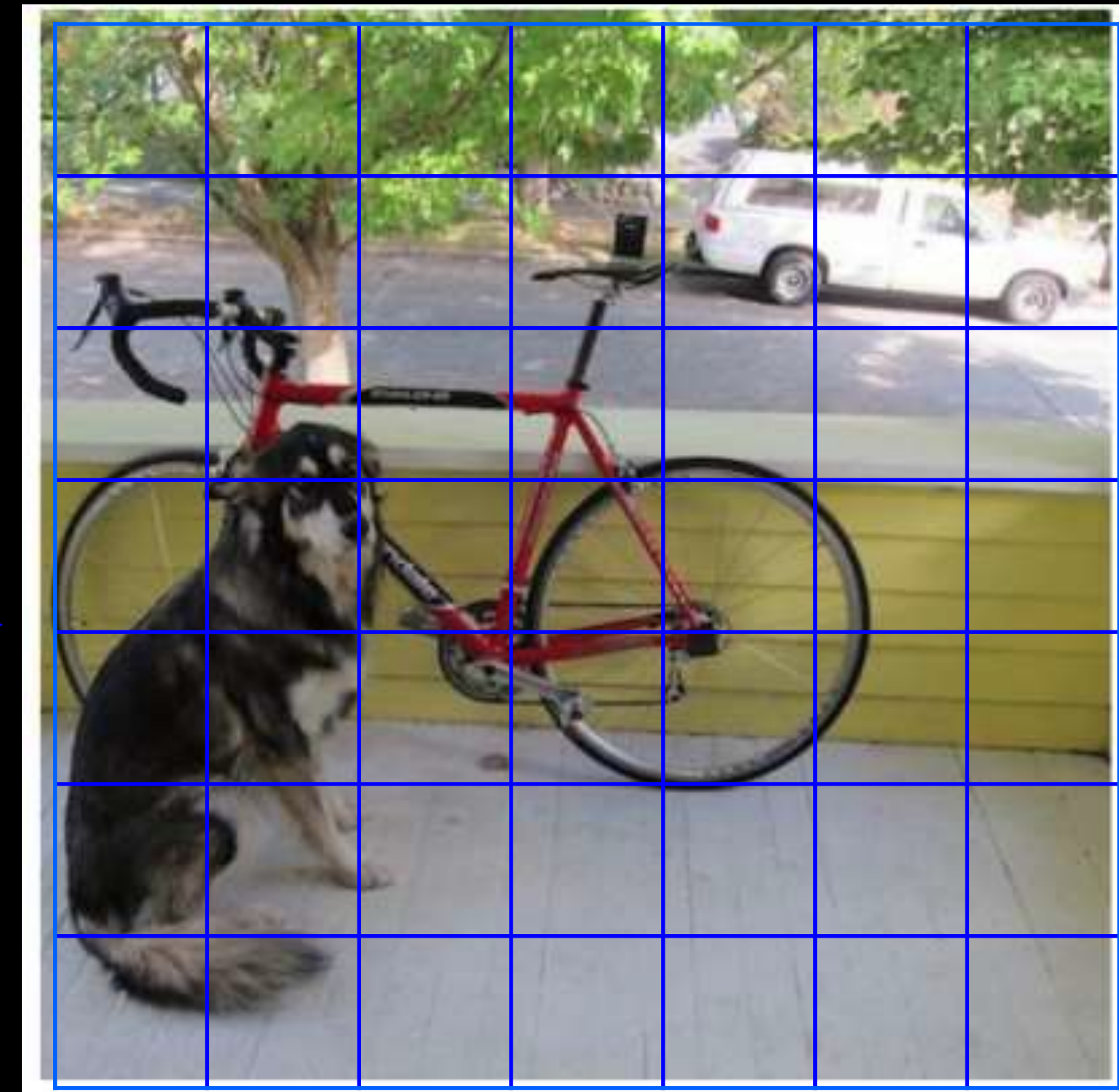
27 October
2025





YOLO formulates object detection as a single regression task — one network predicts all boxes and classes.

Each grid cell makes its own guess: is there an object here, what kind, and where?”



27 October
2025

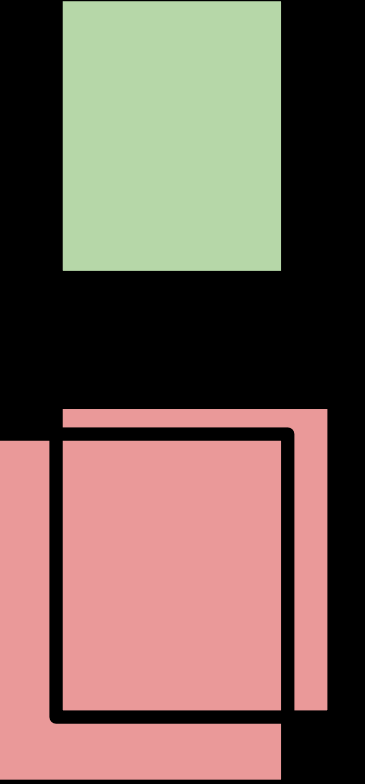




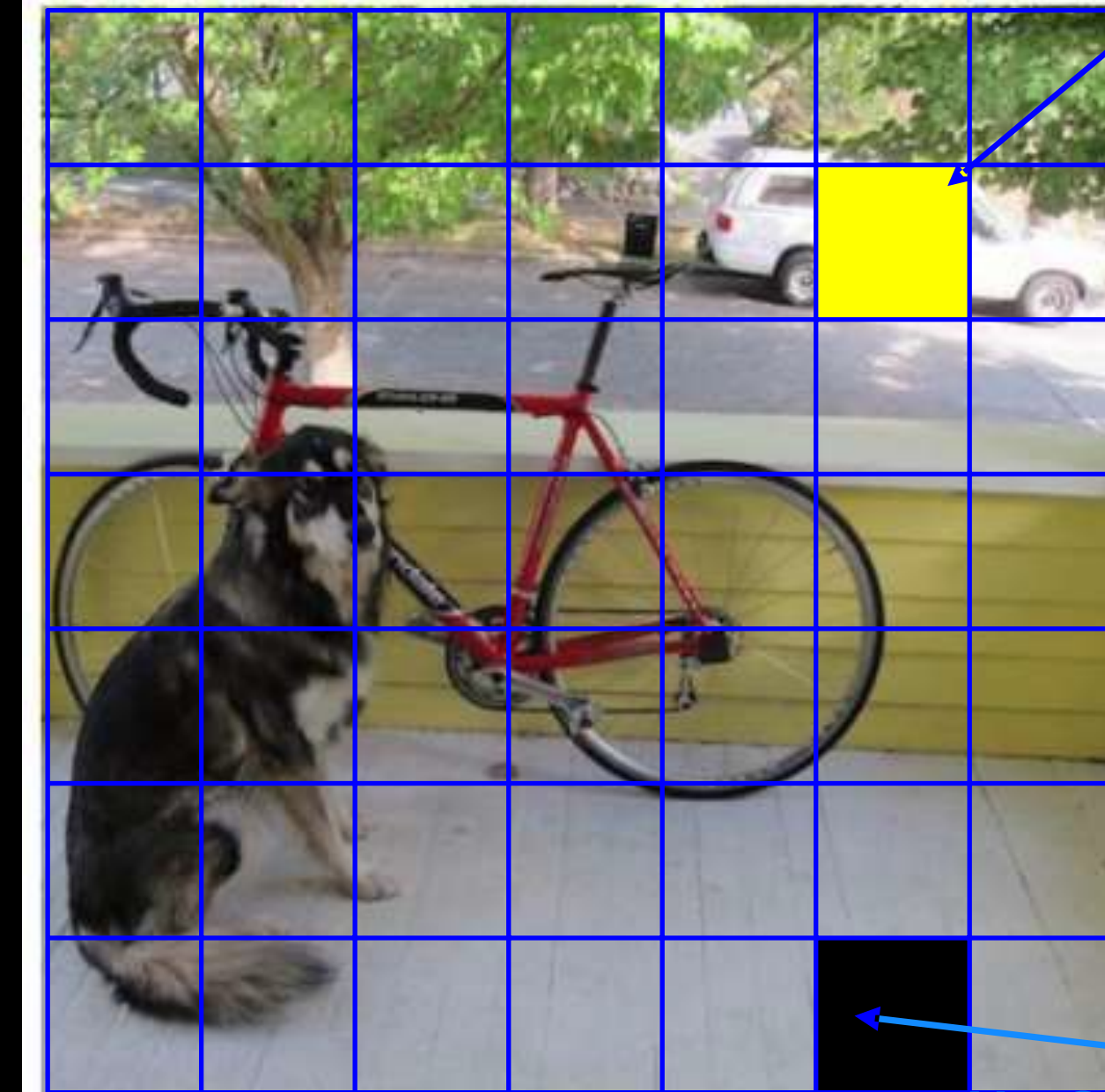
Confidence score

- Each grid cell predicts B bounding boxes and confidence scores for those boxes.
- If a cell has an object, then confidence score = Intersection over union (IOU) between the predicted box and the ground truth.

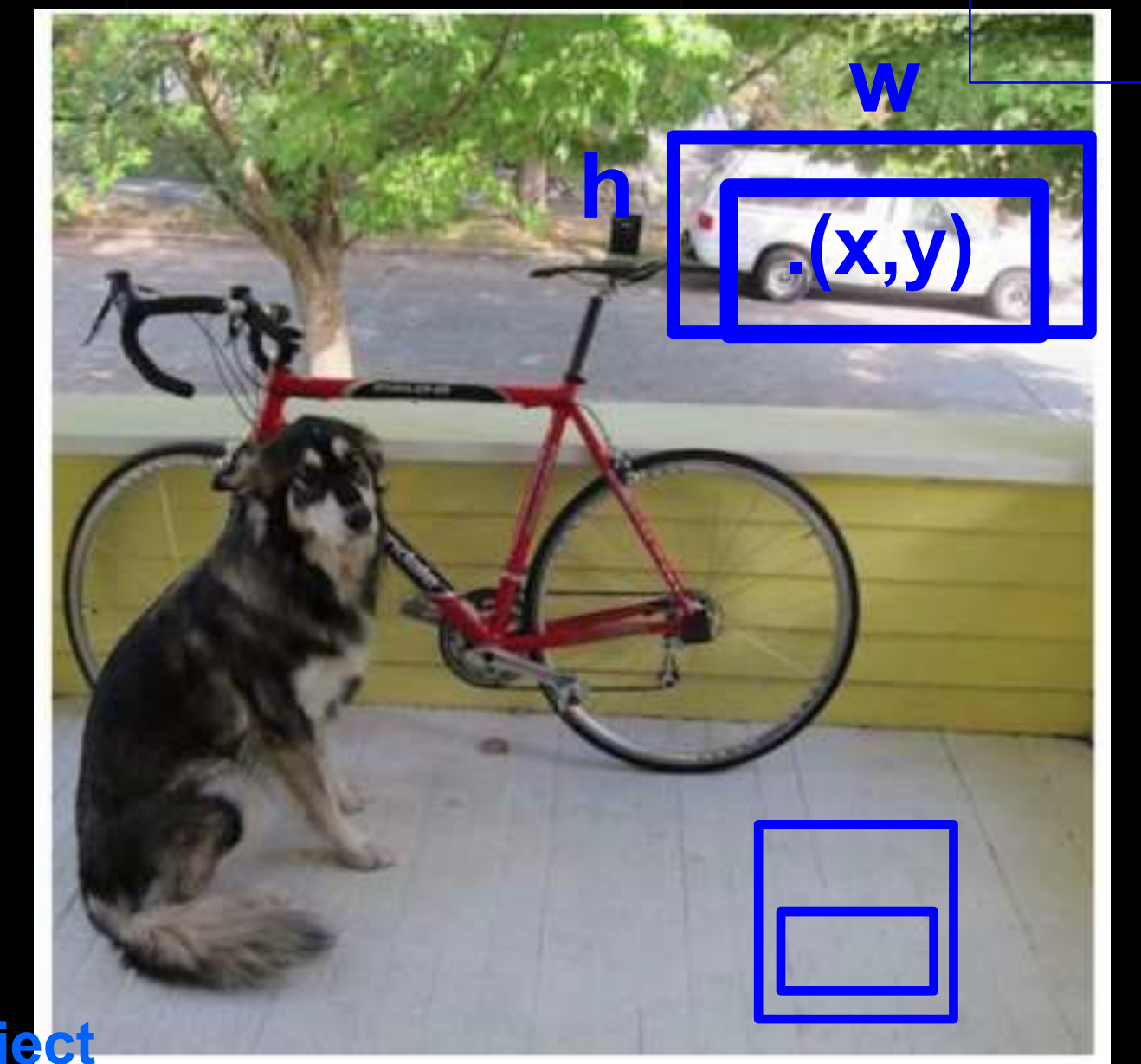
Each cell predicts B boxes (x,y,w,h) and confidences of each box:



$B = 2$



No Object





Each cell also predicts class probability

Car

Car: 0.7
Bicycle : 0
Dog: 0

Bicycle

E.g.

Dog : 0

Car : 0

27 October
2025

Bicycle : 0.7

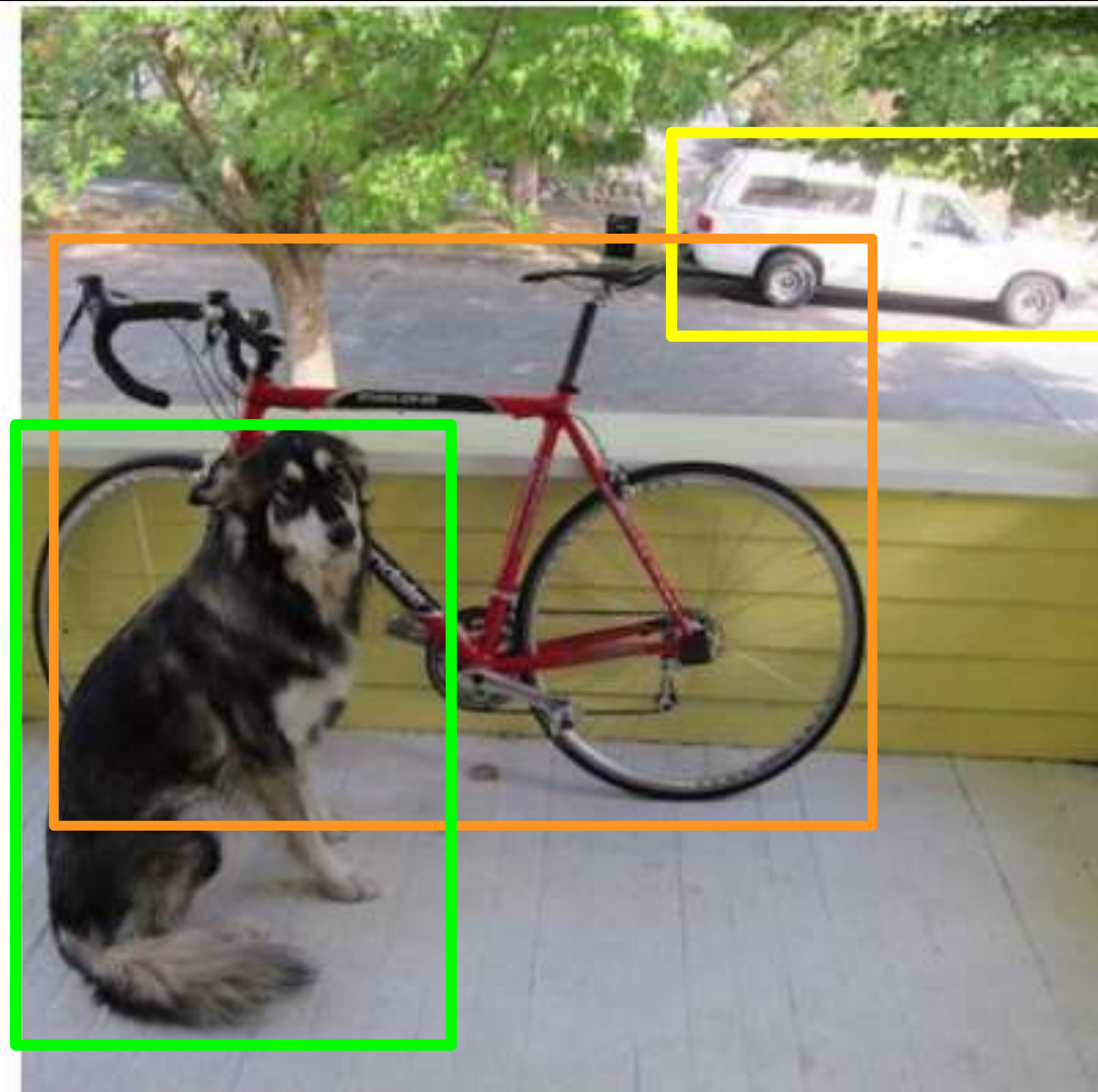
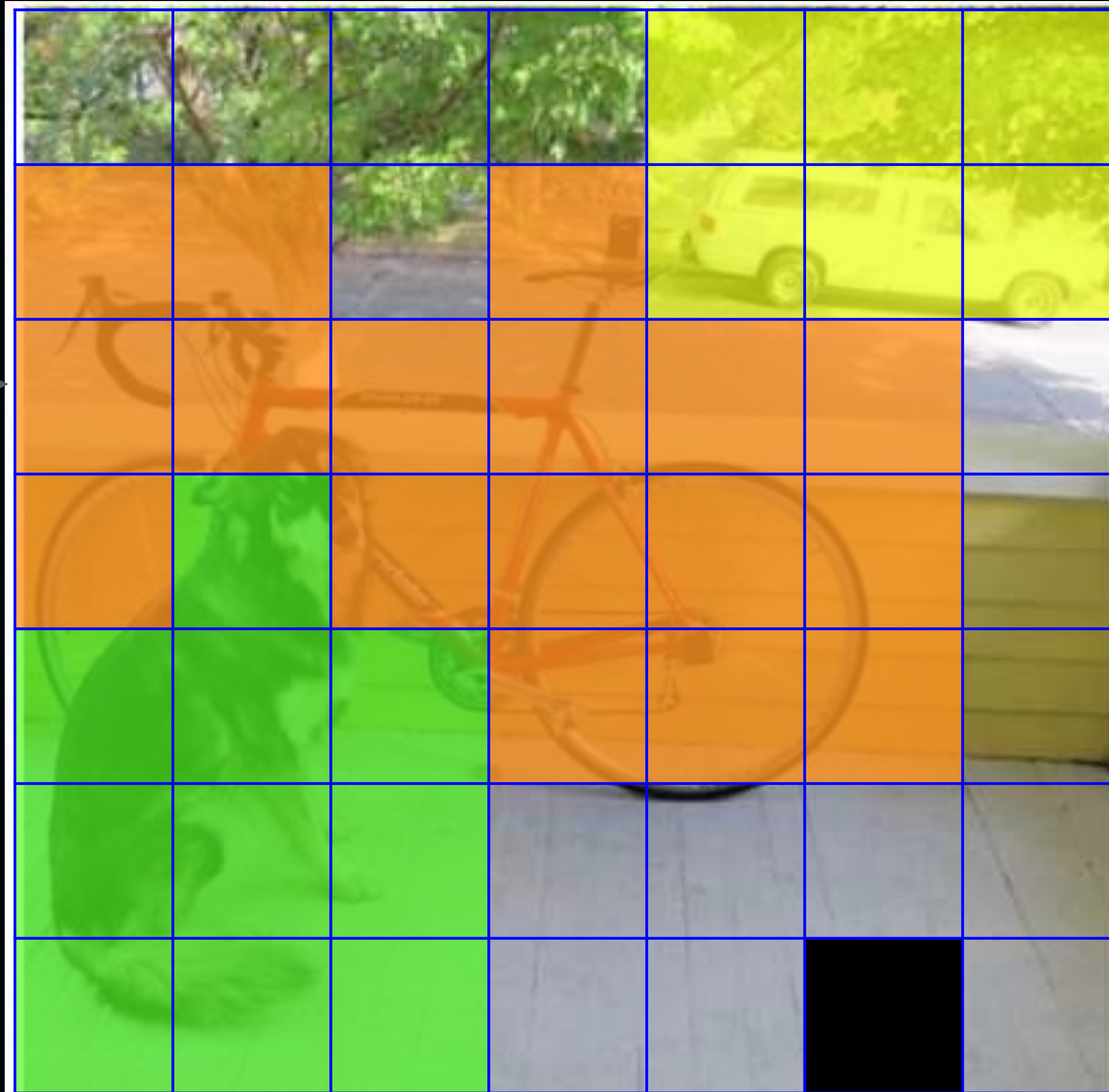
Dog

E.g.

0.8

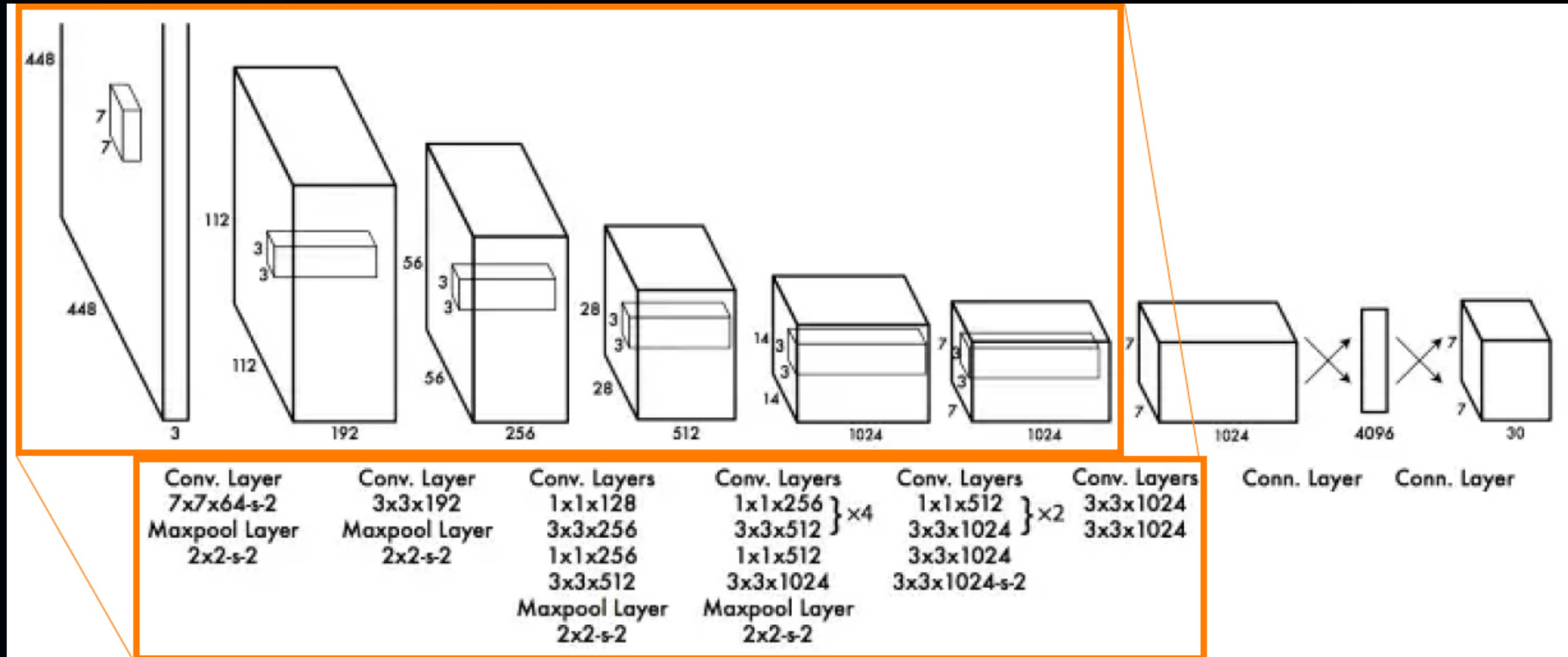
Dog :

Car : 0





Inside YOLO (Network Design)



27 October
2025

- Inspired by the GoogLeNet (image classification)
- 24 convolutional layers followed by 2 fully connected layers
- Fast YOLO uses 9 convolutional layers (instead of 24)





What YOLO Learns (Loss Function simplified)

1. What is a Loss Function?

- A tool to measure how well our model is performing.
- A lower loss indicates better performance, meaning our predictions are close to the actual values.

2. Components of YOLO Loss Function:

- Localization Loss: → teaches the model to draw boxes correctly
- Classification Loss: → teaches to name the right object
- Confidence Loss: → teaches when *not* to predict

3. Balancing the Loss:

- Weights are added to balance the contribution of different components in the loss function.
- For example, YOLO assigns more importance to accurate localization than accurate classification.

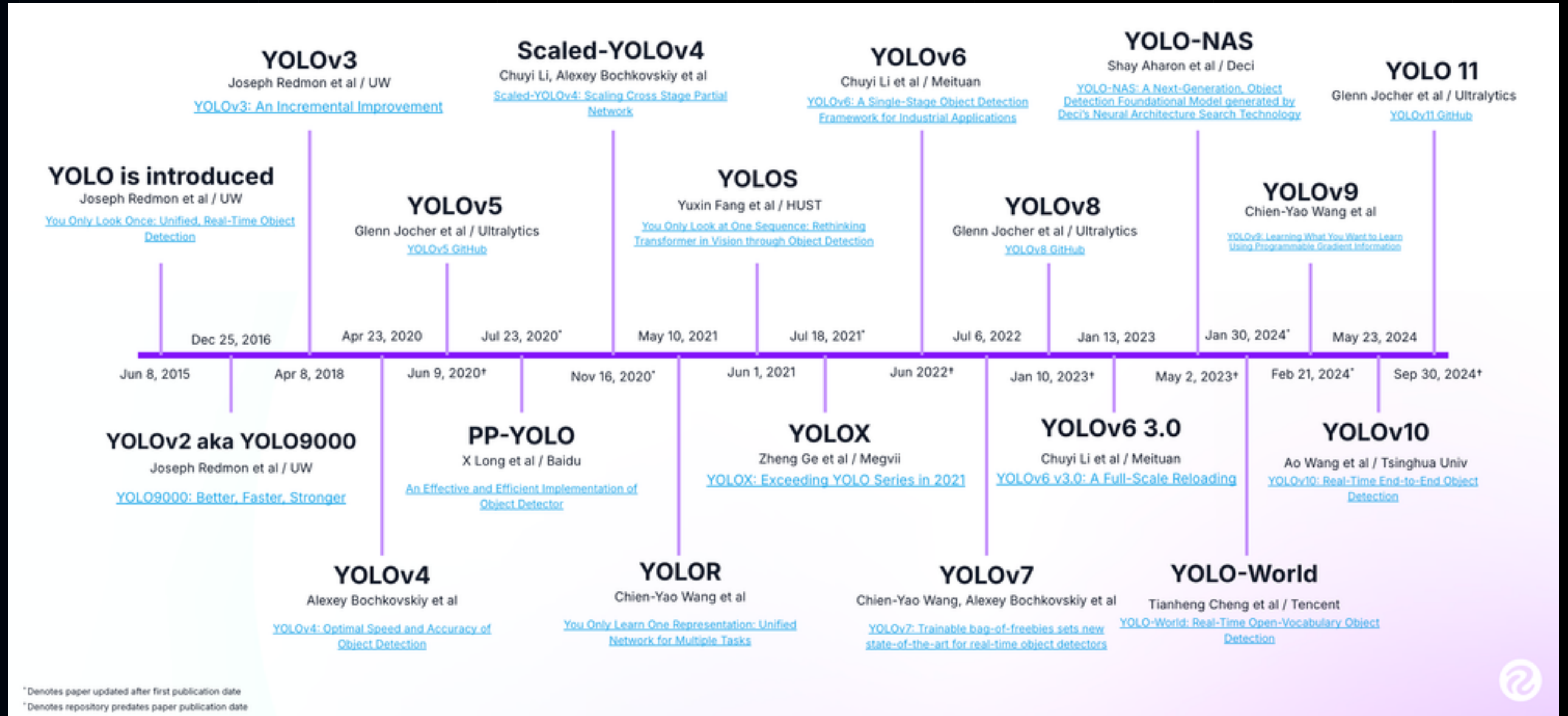
4. The objective of Training:

- The goal is to adjust the model parameters to minimize the loss function.
- A more minor loss means the model's predictions are closer to the truth.





From YOLOv1 to YOLOv9 — faster, anchor-free, and even more precise.



27 October
2025





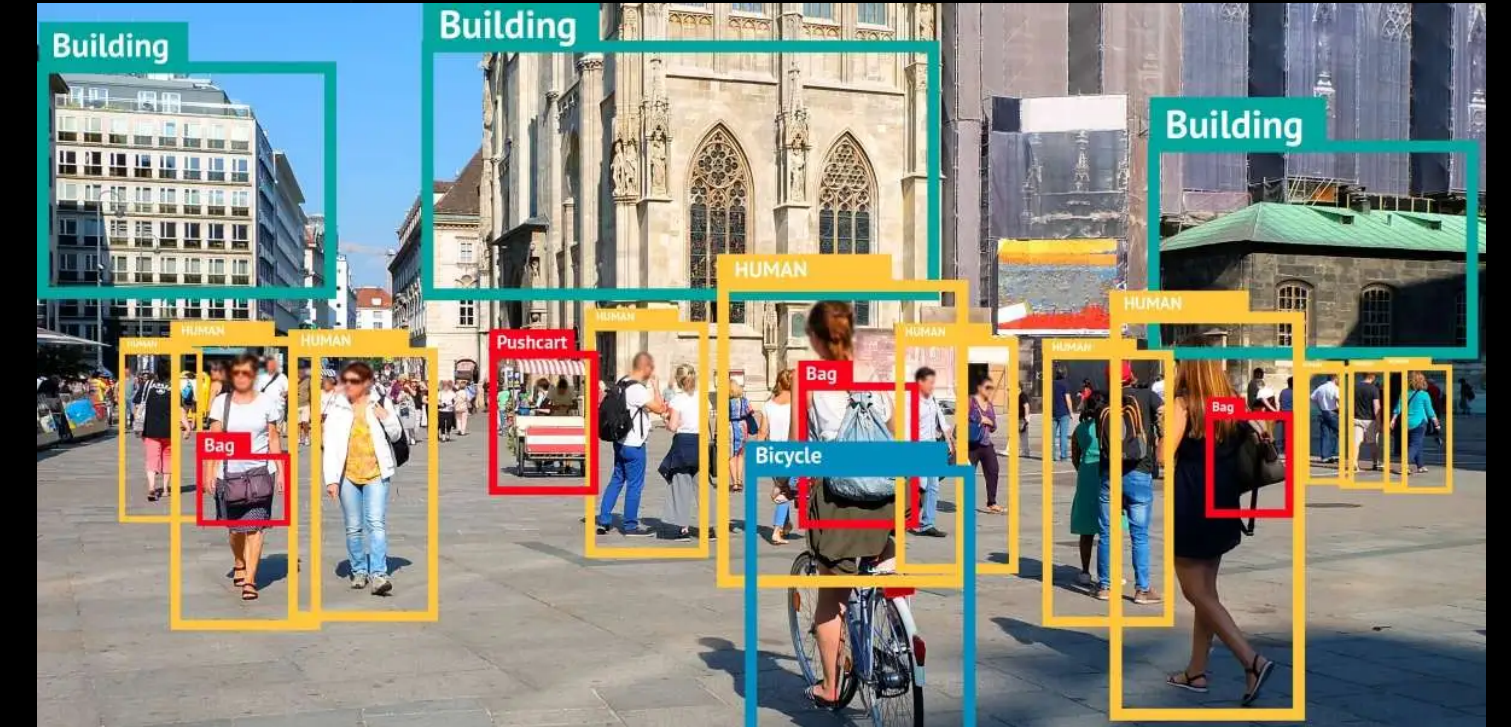
Positives & Limitations of YOLO

Pro:

- Accurate, real-time object detection on natural images
- High inference speed vs. older architectures
- Strong generalization across domains (smaller precision drop than R-CNNs)
- Fewer background errors (4.8% vs. 13.6% in Fast R-CNN)

Cons:

- Each grid cell predicts only two boxes / classes
- Struggles with small or overlapping objects
- Weak on unusual shapes or aspect ratios
- Treats large and small boxes equally in error calculation
- Misses multiple objects in the same cell
- Unstable gradients early in training
- No batch normalization (in early versions)
- Trains at 224×224 , detects at 448×448





Person Detection in Artwork

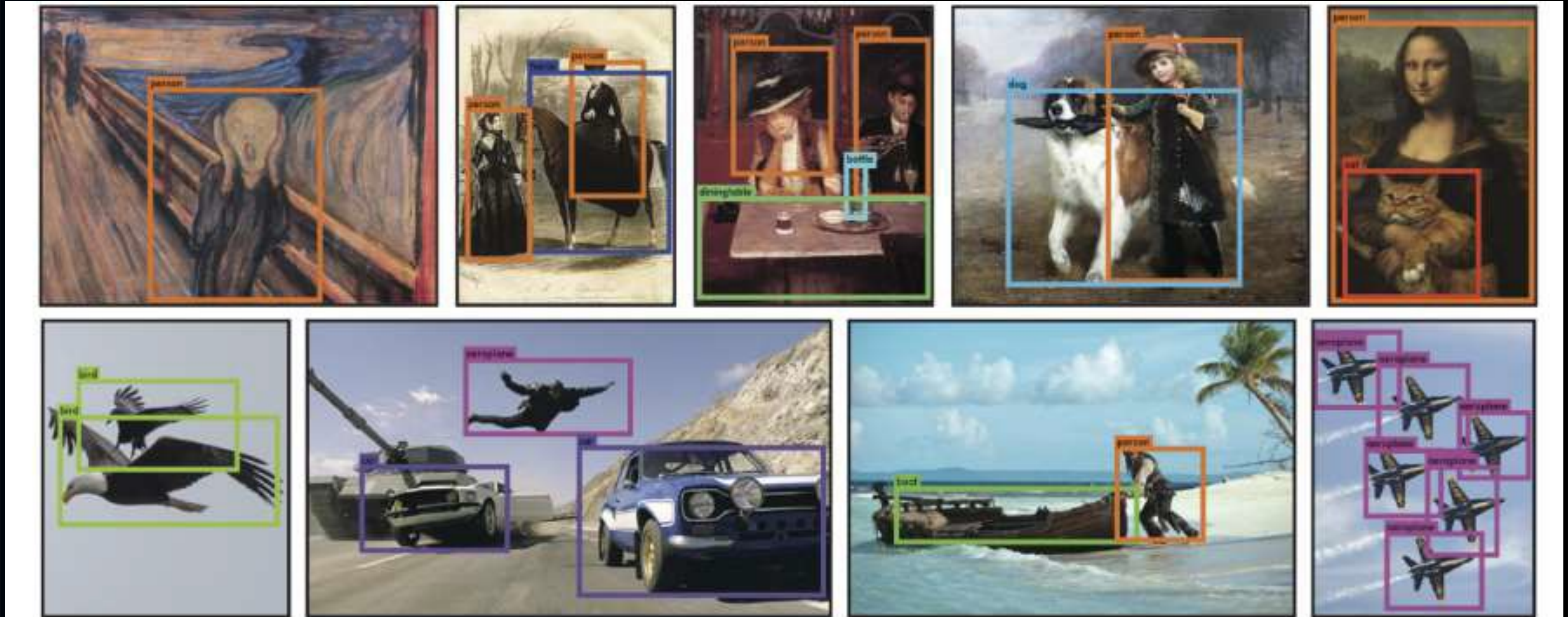


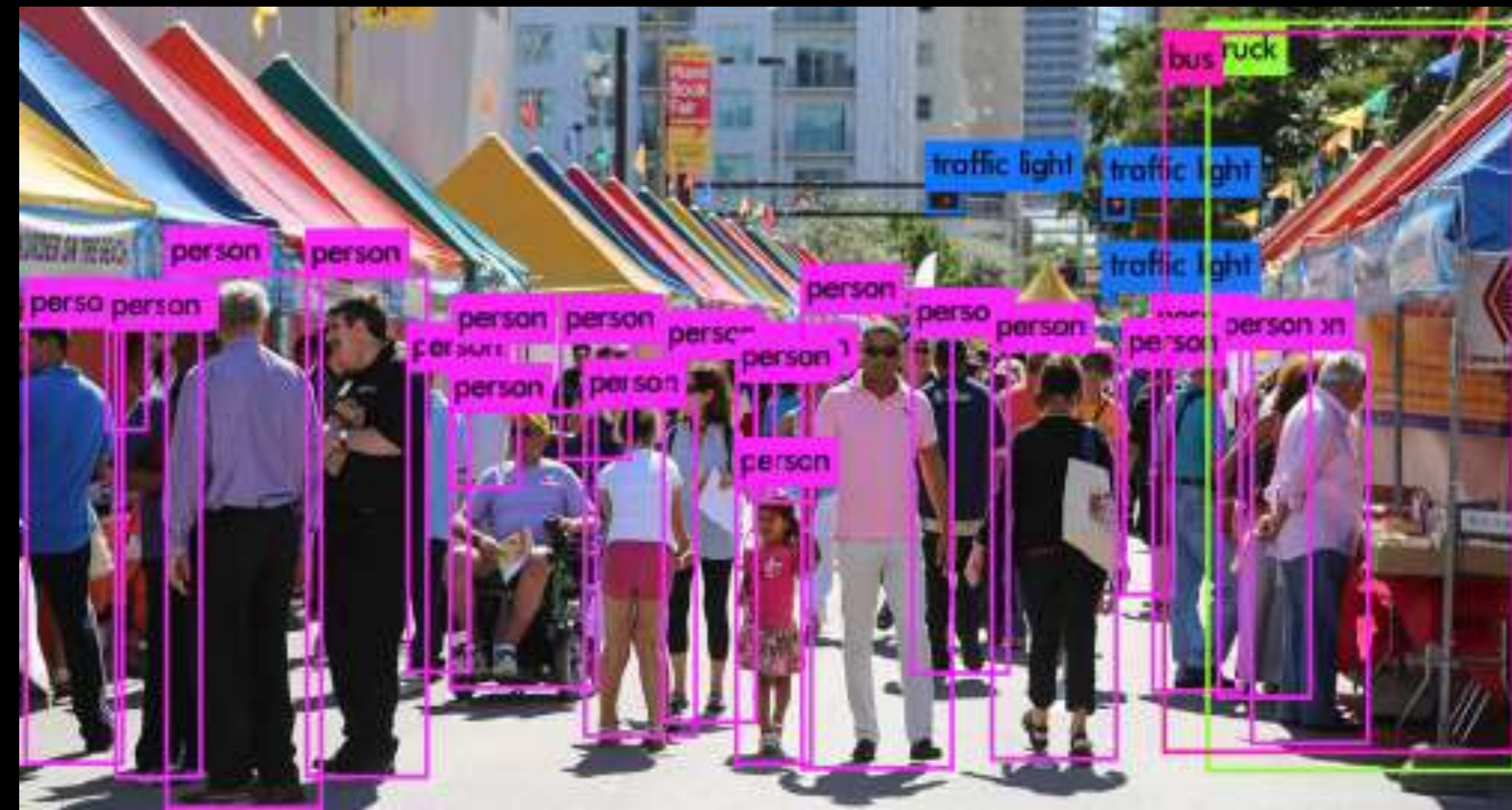
Figure 6: Qualitative Results. YOLO running on sample artwork and natural images from the internet. It is mostly accurate although it does think one person is an airplane.

27 October
2025





Darknet (YOLO) results on random images



27 October
2025





Practical Activity

27 October
2025



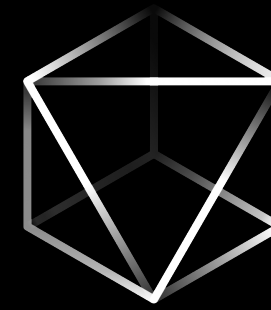


Kick Off Questionnaire

Thanks & Questions?

27 October
2025





Aslihan Yilmaz-Hauck | AI & Computer Vision WS25

HSD Düsseldorf | BMI